10

Systemic Antifungal Agents

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QUESTIONS

Q10.1 What are several of the drugs currently available for systemic fungal indications or in the drug development process that have a future role in dermatologic drug therapy? **(Pg. 100)**

Q10.2 Considering terbinafine, itraconazole, and fluconazole, (1) which drug has the greatest overall bioavailability, and (2) which drug's bioavailability is most affected by gastric pH? **(Pg. 102)**

Q10.3 How do the pharmacokinetics of terbinafine, itraconazole, and fluconazole in (1) the sweat, sebum, and stratum corneum, (2) the nails, and (3) the hair influence the option of intermittent/pulse therapy? **(Pg. 102x3)**

Q10.4 What is a possible explanation for the higher efficacy of terbinafine against endothrix compared with ectothrix infections in tinea capitis? **(Pg. 103)**

Q10.5 Considering the allylamine mechanism (terbinafine), (1) what enzyme is inhibited, (2) what conversion step is inhibited, and (3) is the net in vitro result fungicidal or fungistatic? **(Pg. 103)**

Q10.6 Considering the azole mechanism (itraconazole, fluconazole), (1) what enzyme is inhibited, (2) what conversion step is inhibited, and (3) is the net in vitro result fungicidal or fungistatic? **(Pg. 103)**

Q10.7 Considering all five main systemic antifungal agents in this chapter, which have the most optimal coverage against (1) *Candida* infections, (2) dermatophytes, (3) nondermatophyte mold infections, and (4) *Pityrosporum* infections? **(Pgs. 103, 105, 106)**

Q10.8 Why did terbinafine and itraconazole largely replace griseofulvin as treatment for dermatophyte onychomycosis? **(Pg. 105)**

Q10.9 Considering terbinafine, itraconazole, fluconazole and griseofulvin treatment of tinea capitis in children, what pharmacokinetic properties might influence their efficacy? **(Pg. 106)**

Q10.10 In which clinical situations should oral antifungals (as opposed to topical antifungals) be used for tinea corporis, tinea cruris, and tinea pedis? **(Pg. 106)**

Q10.11 Which antifungal agent has a significant risk of inducing (1) congestive heart failure and (2) depression? **(Pgs. 108, 111, 113)**

Q10.12 What are the rare severe dermatologic adverse events reported for terbinafine, itraconazole, and fluconazole? **(Pgs. 110, 111x2)**

Q10.13 Considering all four main systemic antifungals discussed, which is (1) the strongest cytochrome P-450 (CYP)3A4 inhibitor, (2) a CYP2D6 inhibitor, (3) a CYP3A4 inducer, and (4) a CYP2C9 inhibitor (and the most important drug interactions with each)? **(Pg. 111x2, Tables 10.6 and 10.7)**

Q10.14 Considering all four main systemic antifungal agents in this chapter, which has drug interactions with a risk of (1) torsades de pointes, and (2) inducing an increased international normalized ratio? **(Pg. 111x2)**

ABBREVIATIONS USED IN THIS CHAPTER

AE Adverse event/effect(s)
ALT Alanine aminotransferase
AST Aspartate aminotransferase
CYP Cytochrome P-450
DLSO Distal lateral subungual onychomycosis
FDA US Food and Drug Administration
HIV Human immunodeficiency virus
IV Intravenous
LFT Liver function tests
MIC Minimum inhibitory concentration
OL Off-label use
PSO Proximal subungual onychomycosis
ULN Upper limit of normal
URTI Upper respiratory tract infection

Introduction

The availability of modern antifungal agents has greatly improved the treatment of a wide range of superficial and systemic dermatomycoses. Systemic antifungal agents have been particularly helpful in improving the efficacy of onychomycosis and tinea capitis treatment. Four main systemic agents for superficial dermatologic indications are available: (1) terbinafine, (2) itraconazole, (3) fluconazole, and (4) griseofulvin. This chapter focuses on terbinafine, itraconazole and fluconazole, as the traditional agents, and griseofulvin, which now has limited use in the management of onychomycosis and other dermatomycoses (Table 10.1).[1–9] A new generation of systemic antifungal agents has been developing, including voriconazole, posaconazole, ravuconazole, and albaconazole.[10–12] These agents are not currently approved for any superficial dermatomycoses, but voriconazole and posaconazole have approved systemic antifungal indications and in this chapter will be briefly compared and contrasted with their predecessors, itraconazole and fluconazole.

Terbinafine is an allylamine that was developed in 1974. Oral terbinafine was first approved for use in the United Kingdom in February 1991, in Canada in May 1993, and in the United States in May 1996. Topical terbinafine was approved in the United States in December 1992. The first allylamine to be approved for use in humans was naftifine; however, it is only available as a topical treatment.

Itraconazole is a first-generation triazole that was synthesized in 1980. It first obtained international registration in 1987. Itraconazole was approved in the United States for the treatment of systemic mycoses in September 1992, and as continuous therapy for onychomycosis of the fingernails and toenails in October 1995. Itraconazole pulse therapy has been approved in the United States for the treatment of fingernail onychomycosis since January 1997.

Fluconazole is a first-generation triazole that was initially approved for use in humans in France and the United Kingdom in 1988 and in the United States in January 1990. In September 1993, Finland and China were the first countries to receive approval for fluconazole treatment of onychomycosis. In the United States, fluconazole is not approved for the treatment of onychomycosis or other dermatomycoses.

Griseofulvin was isolated from the mold *Penicillium griseofulvum Dierckx* by Oxford in 1939. In the late 1950s and early 1960s, griseofulvin was found to be effective in the treatment of superficial fungal infections in humans. It was the first significant oral antifungal agent available for the management of dermatomycoses. Although the use of griseofulvin has declined over the years, it is still widely used for the treatment of tinea capitis in children.

Editor's note—the following three paragraphs are left in for historical perspective, but ketoconazole, voriconazole, and posaconazole have virtually no significant role in dermatologic practice today and will not be discussed in the remainder of this chapter.

Ketoconazole was the first significant oral imidazole to become available for the treatment of mycotic infections. The drug was released in the United States in 1981. Previous therapies for systemic mycoses or severe dermatomycoses required intravenous administration (e.g., amphotericin B or miconazole), or had a limited spectrum of action (e.g., griseofulvin for dermatophytes or nystatin for *Candida albicans*).

In July 2013, the US Food and Drug Administration (FDA) announced that it was limiting the use of ketoconazole oral tablets because of hepatotoxicity concerns, adrenal gland problems, and risk of drug interactions. Ketoconazole tablets are not indicated for treatment of fungal infections of the skin and nails and are not indicated for *Candida* and dermatophyte infections.[13,14]

Voriconazole, posaconazole, ravuconazole, and albaconazole are members of a second generation of triazoles. **Q10.1** Voriconazole and posaconazole have been respectively approved for the treatment and prophylaxis of invasive fungal infections, whereas ravuconazole and albaconazole are still under clinical investigation. Posaconazole, ravuconazole, and albaconazole have been used for onychomycosis clinical trials with favorable results, and these new antifungals might have future dermatologic indications.

General Pharmacokinetic Properties of the Oral Antifungals

The major properties of the most frequently used oral antifungals are summarized in Table 10.2.[1–12] Antifungal agents typically fall into the structural families: (1) triazoles—itraconazole, fluconazole, voriconazole, posaconazole, ravuconazole, and albaconazole, (2) imidazoles—ketoconazole, and (3) allylamines—terbinafine, naftifine (Fig. 10.1). Triazoles and imidazoles are subsets

TABLE 10.1 Systemic Antifungal Agents

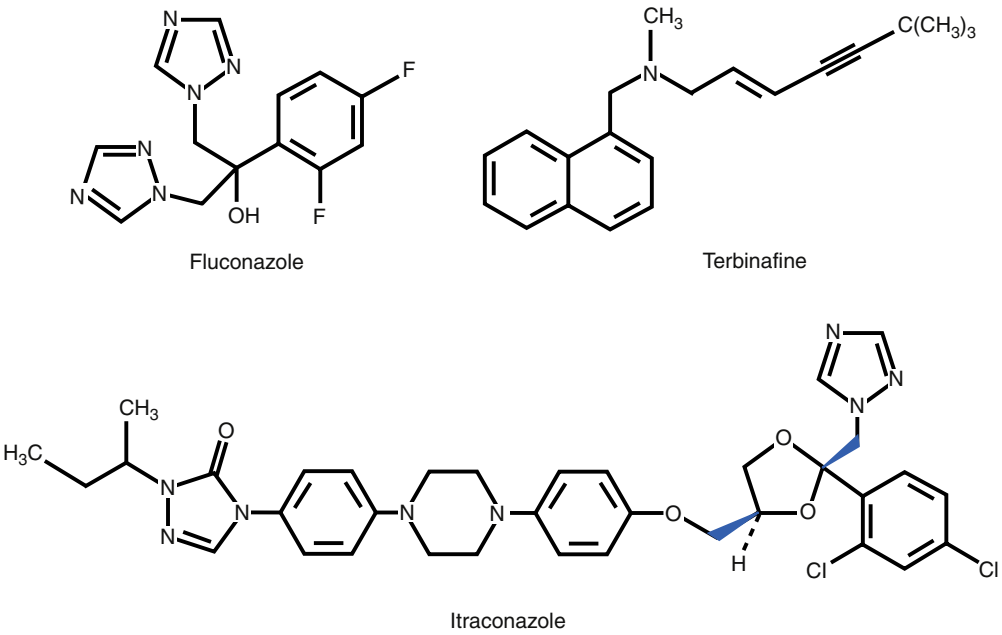
Generic Name	Trade Name	Generic Available	Manufacturer	Dosage Form/Strength	Alternate Formulations
Terbinafine	Lamisil	Yes	Novartis Pharmaceuticals Corp	250 mg tablet	Oral granules 125 mg or 187.5 mg (discontinued)
Itraconazole	Sporanox Onmel	Yes No	Janssen Pharmaceuticals Inc Stiefel Labs	100 mg capsule 200 mg tablet	IV (discontinued), oral suspension 10 mg/mL
Fluconazole	Diflucan	Yes	Pfizer	50, 100, 150, 200 mg tablets	IV 2 mg/mL, oral suspension 50 mg/5 mL, 200 mg/5 mL
Griseofulvin	Gris-PEG	Yes	Pedinol Phamacal	125, 250 mg tablet (ultramicrosize)	
	Grifulvin V	Yes	Ortho Dermatologics	500 mg tablet (microsize)	Oral suspension 125 mg/5 mL

IV, Intravenous; *PEG*, polyethylene glycol.

TABLE 10.2 Key Pharmacology Concepts—Systemic Antifungal Agents

	Terbinafine	Itraconazole	Fluconazole	Griseofulvin
Structure	Allylamine	Triazole	Triazole	Spiro-benzo[*b*]furan
Formulations	Tablet, oral granules	Capsule, oral suspension, IV, tablet	Tablet, oral suspension, IV	Tablet, oral suspension
Peak levels post-dosing	2 hours	3–5 hours	1–2 hours	2-6 hours
Half-life	36 hours (27–30 hours—granules)	21 h (37–40 hours—solution)	30 hours	9-21 hours
Percentage bio-availability	40%[a]—Tablets 36%–64%—Granules	55%	>90%	25-70%
Impact of food	Tablets not affected by food; granules to be sprinkled on food	Capsules: highest levels: full meal or fasted with cola beverage Suspension: take without food	Not affected by food intake	Absorption is significantly enhanced by administration with or after a fatty meal
Protein-binding	>99%	99.8%	11%–12%[b]	84%
Metabolism	Significant first-pass hepatic metabolism; CYP2C9, 1A2, 3A4, 2C8, 2C19 major isozymes; no active metabolites	Extensive metabolism in liver by CYP3A4; active metabolite: hydroxyl-itraconazole	Little first-pass hepatic metabolism, much of dose is excreted as unchanged parent drug	Hepatic, major metabolites are 6-demethyl-griseofulvin and its glucuronide conjugate
Excretion	Renal 70%; clearance decreased 50% in renal impairment or hepatic cirrhosis	Renal 40% inactive metabolites; fecal 3%–18% parent drug	Renal 80% as parent drug, 11% metabolites; 2% feces	Renal 50%, 1% excreted unchanged in urine; 36% in feces

IV, Intravenous.
[a]Nonlinear PK: terminal $t_{1/2}$ is dose-dependent and not predictive of accumulation or elimination.
[b]Fluconazole is much less lipophilic, thus more hydrophilic than other azoles including itraconazole; this accounts for low protein binding.



• **Fig. 10.1** Drug structures—terbinafine, itraconazole, fluconazole.

of the broader category of azole antifungals. Q10.2 Terbinafine and itraconazole have bioavailabilities of approximately 40% and 55%, respectively, and both drugs are extensively metabolized by the liver (see Table 10.2).[1,5] In contrast, fluconazole has a bioavailability of greater than 90%, and undergoes little hepatic metabolism. Absorption of itraconazole in a fasting state, or in individuals with relative or absolute achlorhydria (e.g., those on H_2 inhibitors, antacids, or proton-pump inhibitors), may be increased when the itraconazole is administered with at least 8 oz of a cola beverage.[1] When fluconazole 50 to 400 mg is given once daily, steady-state concentrations are achieved within 5 to 10 days, but may be attained more rapidly by doubling the dose on the first day.[4]

In patients with liver cirrhosis, terbinafine clearance was reduced by approximately 50% compared with normal volunteers, and itraconazole elimination half-life showed a two-fold increase.[1,5] Careful monitoring is suggested in these patients. In patients with renal insufficiency (serum creatinine >3.4 mg/dL, or creatinine clearance <50 mL/min), terbinafine clearance is decreased by about 50%.[5] Itraconazole bioavailability is slightly reduced for subjects with renal insufficiency.[1] Elderly subjects with age-related renal impairment may require dosing adjustments.

Metabolism of oral antifungals by the cytochrome P-450 (CYP) enzyme system provides the potential for interactions with concomitant medications. Drug interactions are discussed later in this chapter.

Pharmacokinetics in Skin

Terbinafine. Q10.3 Terbinafine is delivered to the stratum corneum by passive dermoepidermal diffusion, through sebum and through incorporation of drug from migrating basal keratinocytes.[15] High concentrations of drug (well above the minimum inhibitory concentration [MIC]) are reached in the stratum corneum within hours of starting therapy.[16] Within 2 days of administering terbinafine 250 mg daily, the drug achieves a high level in the sebum.[16] Terbinafine has not been detected in eccrine sweat.[17] The elimination half-life of terbinafine from the stratum corneum and sebum is 3 to 5 days.[16] After administration of terbinafine 250 mg daily for 12 days, drug concentrations above the MIC for most dermatophytes may be present for 2 to 3 weeks after oral therapy is discontinued.[16]

Itraconazole. Itraconazole is delivered to the skin mainly as a result of passive diffusion from the plasma to the keratinocytes, with strong drug adherence to keratin.[18] Itraconazole becomes detectable in the sweat within 24 hours after the initial intake of drug. Despite the early detection in sweat, excretion of itraconazole by this route is minimal, in contrast to griseofulvin, ketoconazole, and fluconazole.[19,20] There is extensive excretion of itraconazole into sebum.[18] A negligible amount of itraconazole redistributes back from the skin and appendages to the plasma; therefore, itraconazole is eliminated as the stratum corneum renews itself, and when the hair and nails grow out.[18,20] Itraconazole may persist in the stratum corneum for 3 to 4 weeks after discontinuation of therapy.[18] In an ex vivo model, the therapeutic effect of itraconazole in the stratum corneum remained for 2 to 3 weeks after stopping therapy.[21]

Fluconazole. When two doses of fluconazole (150 mg once weekly) are administered, the drug accumulates in the stratum corneum through sweat and by direct diffusion through the dermis and epidermis.[22] Excretion in the sebum may be more limited. After discontinuation of therapy, there may be rediffusion of fluconazole from the skin into the systemic circulation, but at a rate lower than elimination from the plasma.[20] In healthy subjects given a single fluconazole dose of 200 mg, the concentration in the plasma and stratum corneum was 3 μg/mL and 98 μg/g, respectively, 7 hours after administration.[23] Over a 5-day fluconazole course (200 mg daily), the elimination of fluconazole from the stratum corneum occurred with a half-life of approximately 60 to 90 hours.[23] This was 2 to 3 times slower than the elimination from plasma. The pharmacokinetic data suggest that fluconazole 150 mg given once weekly should be effective in the treatment of cutaneous fungal infections.

Pharmacokinetics in Nails

Terbinafine. Q10.3 In patients given terbinafine 250 mg daily, the drug may be detected in the distal nail in the majority of patients within 1 week of starting therapy.[17,24,25] In peripheral nail clippings, levels of terbinafine were 0.43 μg/g after 7 days of therapy, or 10 to 100 times the MIC for most dermatophytes. The data indicate that terbinafine diffuses into the nail plate via both the nail matrix and the nail bed. When terbinafine 250 mg daily is administered for 6 or 12 weeks to treat onychomycosis, maximum nail levels of 0.52 and 1.01 μg/g respectively are detectable after 18 weeks of therapy.[24] After completion of 6 and 12 weeks of therapy, terbinafine is detected in the nail for 30 and 36 weeks, respectively.[24]

Itraconazole. Itraconazole has been detected in the distal end of the fingernail and toenail within respectively 1 and 2 weeks of starting therapy.[26,27] This is consistent with itraconazole reaching the free end of the nail plate via both the nail matrix and the nail bed. Furthermore, after receiving itraconazole 200 mg daily for 10 days, the concentration of itraconazole in the subungual nail material is more than double the corresponding level in the distal nail clippings. In patients treated with pulse therapy, 200 mg twice daily for 1 week, the drug levels achieved in the fingernails exceed the MIC.[20] After itraconazole intermittent therapy for 2 pulses (200 mg twice daily for 1 week and 3 weeks off drug), the drug becomes undetectable in fingernail by 9 months after the initiation of therapy.[20] When toenail onychomycosis is treated with 3 pulses of itraconazole, drug levels have been detected in the nail for 11 months.[20] The faster clearance of the drug from fingernails than from toenails is probably because of the faster outgrowth of the former.[27,28] In contrast, plasma levels decrease to very low or negligible levels within 7 to 14 days of stopping itraconazole therapy.

Fluconazole. Fluconazole demonstrates rapid uptake to the nail and may be detected at the distal end of the nail plate within 1 day of starting treatment with 50 mg.[29] This suggests that fluconazole diffusion from the nail bed to the nail plate is an important route of drug delivery. When 150 mg of fluconazole is given once weekly for 12 months, it can be detected in the nail plate for at least 6 months after the discontinuation of therapy in both healthy and diseased nails (n = 36 patients).[30] After 1 and 6 months of therapy, the mean concentration of fluconazole in healthy nails was 3.09 and 8.54 μg/g, respectively. Fluconazole concentrations in healthy and diseased toenails were 1.4 and 1.9 μg/g, respectively, 6 months after discontinuation of the drug (n = 3 patients). These data suggest that there is the potential for ongoing improvement in the onychomycosis, well after discontinuation of active drug therapy.

Pharmacokinetics in Hair

Terbinafine. Q10.3 Terbinafine has been detected in hair within 1 week of starting therapy.[17] Early detection in hair may be because of delivery of drug to hair via the sebum. The drug may become incorporated into hair by hair matrix cells. When terbinafine

250 mg daily is administered for 14 days, the drug has been detected in hair for at least 50 days after the start of dosing.[24] **Q10.4** It should be noted that terbinafine may be more effective against tinea capitis caused by endothrix organisms (e.g., *Trichophyton tonsurans*) than against ectothrix infections (e.g., *Microsporum canis*), as the drug accumulates preferentially in the hair shaft.[31,32]

Itraconazole. Itraconazole may be delivered to the hair primarily by two routes: via the sebum, and by incorporation into the hair follicle.[33] Preliminary testing of both continuous and pulse regimens of itraconazole found that itraconazole was detectable in hair after 1 week of therapy.[18,34] When a pulse regimen was used for Majocchi's granuloma, the concentration of itraconazole in hair was 2.6- and 3.4-fold higher after the second and third pulses, respectively, than at the end of the first pulse. After discontinuation of pulse therapy, drug was detectable in hair for up to 9 months, suggesting that the pulse regimen with itraconazole may be an option for the treatment of tinea capitis.[34]

Fluconazole. Relatively little has been published about the pharmacokinetics of fluconazole in hair. When fluconazole 100 mg was administered daily to healthy volunteers for 5 days, fluconazole was detectable in scalp hair 4 to 5 months after completion of therapy.[23]

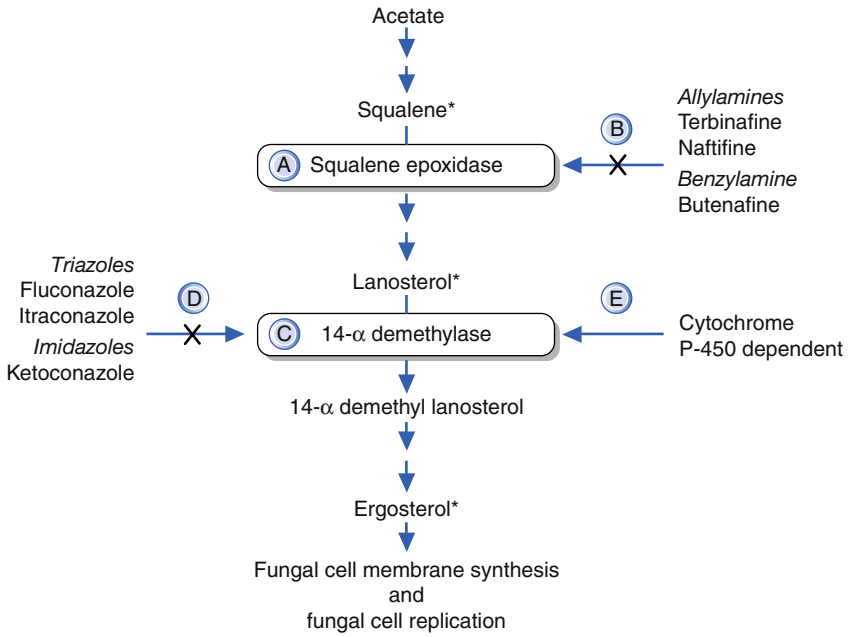
Mechanism of Action

The oral antifungals exert their effect by interfering with the enzymes involved in the manufacture of ergosterol, a crucial component of the fungal cell membrane. Deficiency of ergosterol interferes with membrane function, leading to an arrest in cell growth, which may be fungicidal or fungistatic. It is not clear whether the in vitro fungicidal or the in vitro fungistatic properties of these drug categories translate into similar properties in vivo.

Allylamines—Terbinafine. **Q10.5** Terbinafine and other allylamines inhibit squalene epoxidase, leading to accumulation of squalene and a subsequent deficiency of ergosterol (Fig. 10.2).[35] This deficiency of ergosterol leads to the fungistatic action that is associated with terbinafine use. The accumulation of squalene may be associated with the fungicidal action of terbinafine noted in vitro, possibly by its deposition in lipid vesicles, which leads to cellular membrane disruption.[35]

Triazole—Itraconazole, Fluconazole. **Q10.6** The triazole family inhibits the CYP-dependent enzyme lanosterol 14α demethylase, with resultant inhibition of the conversion of lanosterol to ergosterol (see Fig. 10.2).[10,11,36,37] Because of this mechanism of inhibition, these drugs are associated with a fungistatic action. Azole-resistant organisms have been detected in rare cases following prophylaxis or long-term therapy, and there may be some potential for cross-resistance between azole agents, which has not been adequately characterized.[1,10–12]

Sensitivities of Specific Fungi. **Q10.7** The main *Trichophyton* species, *T. rubrum*, *T. mentagrophytes*, and *T. tonsurans*, have similar low in vitro MIC values to terbinafine and itraconazole.[38] Similarly, low MIC values were found for terbinafine and itraconazole when tested in vitro with *Microsporum* species and *Epidermophyton floccosum*. Fluconazole was less effective than terbinafine or



* Denotes two of the most important precursors (squalene, lanosterol) leading to formation of ergosterol, an essential component of all fungal cell membranes.

A Squalene epoxidase converts squalene to precursors of lanosterol.

B Both allylamine (**terbinafine, naftifine**) and benzylamine (**butenafine**) antifungal agents inhibit the enzyme squalene epoxidase, leading to accumulation of squalene metabolites (fungicidal *in vitro*).

C 14-α demethylase converts lanosterol to 14-α demethyl lanosterol.

D Both triazole (**itraconazole, fluconazole**) and imidazole (**ketoconazole**, others) antifungal agents inhibit 14-α demethylase, leading to ↑ lanosterol and ↓ ergosterol synthesis (fungistatic *in vitro*).

E The enzyme 14-α demethylase is cytochrome P-450 dependent, at least in part explaining the role of the triazoles and imidazoles as "enzyme inhibitors" in the phase I drug metabolism systems.

• **Fig. 10.2** Systemic antifungal agents mechanism.

TABLE 10.3 Systemic Antifungal Agents—US Food and Drug Administration-Approved Indications and Off-Label Uses (Superficial Mycoses)

Condition	Terbinafine	Itraconazole	Fluconazole	Griseofulvin
Onychomycosis: Dermatophyte	FDA-A	FDA-A	OL	FDA-A
Onychomycosis: *Candida* spp.	OL	OL	OL	
Onychomycosis: Nondermatophyte molds	OL	OL		
Tinea corporis/cruris	OL	OL	OL	FDA-A[c]
Tinea pedis	OL	OL	OL	FDA-A[c]
Tinea capitis	FDA-A[a]	OL	OL	FDA-A[c]
Majocchi's granuloma	OL	OL		
Tinea imbricata	OL	OL		OL
Pityriasis (tinea) versicolor		OL	OL	
Pityrosporum folliculitis		OL		
Seborrheic dermatitis	OL	OL		
Chronic mucocutaneous candidiasis		OL	OL	
Vaginal candidiasis		OL	FDA-A	
Cutaneous candidiasis		OL	OL	
Oropharyngeal candidiasis		FDA-A[b]	FDA-A	
Esophageal candidiasis		FDA-A[b]	FDA-A	

[a]Oral granule formulation only.
[b]Oral solution only.
[c]Not justified for minor or trivial infections, which will respond to topical agents alone.
FDA-A, US Food and Drug Administration-approved; *OL*, off-label uses (supported by significant clinical data).

itraconazole against dermatophytes in vitro. However, in vitro testing has confirmed that fluconazole may be effectively used against the dermatophyte organisms. Itraconazole and fluconazole were more effective than terbinafine during in vitro MIC testing of *C. albicans*, and *C. parapsilosis*.[38] Terbinafine and itraconazole showed the lowest MIC values in vitro for nondermatophyte organisms tested, including *Fusarium proliferatum*, *Fusarium solani*, *Onychocola canadensis*, *Scopulariopsis brevicaulis*, *Aspergillus sydowii*, and *Aspergillus terreus*.[38] As expected, the second-generation triazoles, voriconazole and posaconazole, had respectively lower MIC values than fluconazole and itraconazole for dermatophytes and *Candida* species.[38–40] In contrast, ravuconazole had similar MIC values to itraconazole for dermatophytes but similar or lower MIC values than posaconazole for *Candida* species tested.[38]

Clinical Use

US Food and Drug Administration-Approved Indications. Tables 10.3 and 10.4 summarize the indications, both FDA approved and off label, for each of the main oral antifungal drugs.

Terbinafine. Terbinafine is indicated by the FDA for the treatment of dermatophyte onychomycosis of the toenails and/or fingernails in adults.[5] A granule formulation of terbinafine is also approved for tinea capitis in patients 4 years and older.[6]

Itraconazole. Itraconazole is indicated by the FDA for the treatment of dermatophyte onychomycosis of the toenails and/or fingernails in nonimmunocompromised adults. It is also approved in immunocompromised and nonimmunocompromised patients for systemic mycoses, such as blastomycosis, histoplasmosis, and aspergillosis in patients who are intolerant of or refractory to amphotericin B.[1–3]

Fluconazole. Fluconazole is indicated for the treatment of vaginal candidiasis, oropharyngeal and esophageal candidiasis, and cryptococcal meningitis.[4] With regard to prophylaxis, fluconazole is also indicated for use in reducing the incidence of candidiasis in patients undergoing bone marrow transplantation, who receive cytotoxic chemotherapy or radiation therapy.

Griseofulvin. Griseofulvin is indicated for the treatment of dermatophyte infections of the skin, scalp, and nails.[9] As with other oral antifungal agents, the use of griseofulvin is not justified for the treatment of tinea infections that would be expected to respond satisfactorily to topical antifungals. Griseofulvin is not effective in the treatment of pityriasis (tinea) versicolor, bacterial infections, candidiasis (moniliasis), or deep mycotic infections.

Specific Dermatologic Indications (FDA Approved and Off Label)

Dermatophyte Onychomycosis. Resolution of dermatophyte onychomycosis (*Trichophyton* spp., *Microsporum* spp., and *Epidermophyton* spp.) typically requires oral therapy. Limited success has been noted with topical monotherapies, but these treatments are beyond the scope of this chapter.[41] Successful therapy may also require implementation of preventative measures by the patient, which include proper nail care, use of proper footwear, and drying the feet completely after a bath or shower.[42]

TABLE 10.4 Systemic Antifungal Agents—US Food and Drug Administration-Approved Indications and Off-Label Uses (Deep or Systemic Fungal Infections)

Condition	Terbinafine	Itraconazole	Fluconazole
Chromo blastomycosis	OL		
Blastomycosis		FDA-A	
Histoplasmosis		FDA-A	
Aspergillosis	OL	FDA-A (intolerant/ refractory to amphotericin B)	
Sporotrichosis	OL (cutaneous)		OL (lymphocutaneous and visceral)
Systemic *Candida* infection			FDA-A: Prophylaxis (transplant patients receiving chemotherapy and/or radiation therapy)
Other uses	OL: Black piedra	OL: HIV-1 associated eosinophilic folliculitis	FDA-A: Cryptococcal meningitis

FDA-A, US Food and Drug Administration-approved; *HIV*, human immunodeficiency virus; *OL*, off-label uses (supported by significant clinical data).

Use of combinations of antifungals (oral/oral or topical/oral) has had limited testing and will not be discussed further in this chapter. The combination of oral antifungals with differing mechanisms of action may provide synergistic effects upon nail fungal organisms.[43,44] Similarly, adjunct use of a topical medication with an oral drug may reduce the amount of oral drug required, improving the safety and cost of treating severe cases if onychomycosis.[41–43,45] We refer the reader to Baran and colleagues, 2008[46] for review.

Griseofulvin. **Q10.8** Although griseofulvin is approved for tinea infection of the nails, its affinity for keratin is low and long-term therapy (at least 9–12 months) is required.[47] Treatment efficacy is also low, and the newer azole and allylamine agents have largely replaced griseofulvin for this indication.[47]

Terbinafine. For terbinafine, the recommended dosage is 250 mg daily for 6 and 12 weeks for fingernail and toenail onychomycosis, respectively.[42,47] An analysis of studies where terbinafine 250 mg daily was administered for 12 to 16 weeks to treat dermatophyte onychomycosis of the toenails, demonstrated a mycologic cure rate meta-average ± 95% confidence interval (95% CI) of 76% ± 3% (18 trials, n = 993 patients).[48] The clinical response (nail visually clear of infection or showed marked improvement) meta-average rate was 66% ± 5% (15 trials, n = 1199 patients). Similar mycologic cure rates are found for dermatophyte fingernail onychomycosis.[49,50] Several intermittent terbinafine regimens aimed to reduce cost and increase safety are also reported in the literature, albeit not formally approved.[51]

Itraconazole. The FDA-approved regimen for dermatophyte toenail onychomycosis in immunocompetent patients, with or without fingernail involvement, is continuous itraconazole 200 mg daily for 12 weeks.[47] When only fingernails are infected, pulse itraconazole dosing is the approved regimen. Each pulse consists of 200 mg twice daily (400 mg daily) for 1 week per month, with 2 pulses being the accepted regimen.[47] In many countries, pulse therapy is the approved regimen for toenail onychomycosis using 3 to 4 pulses, and this regimen has become the standard in the United States for many physicians despite FDA labeling.[3,47,48]

Where itraconazole was used continuously for 12 to 16 weeks to treat dermatophyte toenail onychomycosis, a mycologic cure rate meta-average ± 95% CI of 59% ± 5% (7 trials, n = 1131 patients) was found.[48] The clinical response (nail visually clear of infection or showed marked improvement) meta-average rate was 70% ± 5% (7 trials, n = 1135 patients). The mycologic cure rate meta-average ± 95% CI for itraconazole pulse therapy for toenails was 63% ± 7% (6 trials, n = 318 patients).[48] The clinical response (nail visually clear of infection or showed marked improvement) meta-average rate was 70% ± 11% (6 trials, n = 329 patients).[48] Itraconazole pulse therapy has also shown good efficacy in small trials for dermatophyte fingernail onychomycosis (2 trials, n = 38 patients).[52,53]

Fluconazole. In countries where fluconazole is approved for the treatment of onychomycosis, the most frequently used schedule is fluconazole 150 to 300 mg once weekly until the abnormal-appearing nail has grown out. This may occur in 3 to 6 months, and 9 to 12 months, for dermatophyte fingernail and toenail onychomycosis, respectively.[47] For toenail onychomycosis, the mycologic cure rate meta-average ± 95% CI was found to be 48% ± 5% (3 trials, n = 131 patients).[48] The clinical response (nail visually clear of infection or showed marked improvement) meta-average rate was 45% ± 21% (3 trials, n = 132 patients). For fingernail onychomycosis, fluconazole showed a mycologic cure rate of 89% to 100% and a clinical cure rate of 76% to 90%.[54]

Onychomycosis Caused by Candida Species and Nondermatophyte Molds. **Q10.7** Nondermatophyte molds and *Candida* species are rarely diagnosed as causative organisms in onychomycosis infections. *Candida* nail infections are associated with immunocompromised status (chronic mucocutaneous candidiasis, human immunodeficiency virus [HIV] infection, use of immunosuppressive drugs).[55] Most oral antifungals have a wide spectrum of action and show efficacy in treatment of nondermatophyte mold onychomycosis, although treatment efficacy may be species specific.[41,55]

Terbinafine and Itraconazole. Data regarding the use of terbinafine to treat nondermatophyte molds are limited relative to dermatophytes.[56] *Aspergillus* spp. infections may be treated effectively with terbinafine or itraconazole.[56] Terbinafine and itraconazole monotherapies may have limited efficacy in many patients with *S. brevicaulis* and *Fusarium* spp. toenail infections, although one study reported a good cure rate with terbinafine for *S. brevicaulis*.[57] Combination of oral therapy with topical therapy may increase treatment efficacy.[55] Oral therapies are reported to be ineffective for *Scytalidium* spp. and *O. canadensis* infections.[55]

Compared with its efficacy against dermatophytes, terbinafine is less effective when the causative agent of the onychomycosis is *Candida* species, in particular *C. albicans*.[58–60] Itraconazole continuous regimens of 200 mg daily or pulse regimens of 400 mg

daily for 1 week each month have been effective against *Candida* onychomycosis.[55,61–63]

Fluconazole. Experience with fluconazole for the treatment of onychomycosis caused by nondermatophyte molds is limited. Fluconazole 150 mg daily for 12 weeks has been effective against *S. brevicaulis*,[57] whereas 50 mg daily or pulse therapy of 300 mg/week has been effective against *Candida* onychomycosis, using a 6-week duration of therapy for fingernails and 3 months for toenails.[55,56]

Tinea capitis. Oral antifungals have been of great benefit in the effective treatment of tinea capitis, as they are able to penetrate the infected hair shaft where topical therapies cannot, thereby providing better cure rates. However, it should be noted that topical therapies, such as ketoconazole shampoo, selenium sulfide, and povidone-iodine can also have an important role as adjunct therapy for tinea capitis infection.[64,65] Control of infection transmission via fomites and symptom-free carriers may need to be addressed, using adjunct therapies and techniques, such as counseling, about not sharing items, such as hats, combs, and toys, and identifying infected pets.[64,65] Griseofulvin and terbinafine are the only antifungals approved for use in tinea capitis. However, terbinafine seems to be superior for the treatment of *T. tonsurans* infection, whereas griseofulvin remains the treatment of choice for *M. canis* infections.[66,67] **Q10.9** This observation may reflect the different infection patterns of the organisms: *Trichophyton* infections are endothrix (inside the hair shaft, where drug accumulation may help eradicate infection), whereas *Microsporum* are ectothrix (outside the hair shaft, requiring drug to be secreted to the hair surface via sweat or sebum). Because children have little sebum secretion before puberty, terbinafine may not reach ectothrix infections adequately for treatment, whereas its incorporation into the hair shaft is effective in endothrix infections.[68] In contrast, griseofulvin and the azoles tend to be secreted in sweat and may reach the ectothrix infections more effectively than terbinafine.[68] The efficacies and safety profiles of terbinafine, itraconazole, and fluconazole are similar to those of griseofulvin, and may be used where griseofulvin therapy has failed.

Griseofulvin. Griseofulvin dosing for the treatment of tinea capitis ranges from 10 to 25 mg/kg daily, based on drug formulation (microsize or ultramicrosize oral suspension), with a treatment duration of at least 6 weeks being recommended for either formulation (Table 10.5).[64,65] One study has suggested that griseofulvin doses >20 mg/kg/day do not significantly increase the complete cure rate compared with less than 20 mg/kg/day.[67] Griseofulvin appears to provide superior efficacy to terbinafine, when treating tinea capitis, because of *Microsporum* sp.[66,67] In some countries, including the United States, griseofulvin oral suspension, 125 mg/5 mL, is available (see Table 10.1). Alternatively, the tablets can be pulverized and administered with food.

Terbinafine. Before 2007 terbinafine tablets were commonly prescribed off label for tinea capitis infections using a weight-based dosing regimen (see Table 10.5).[64,65] In 2007 the FDA approved a terbinafine oral granule formulation for tinea capitis. The label-indicated weight-based dose for the granule formulation is: less than 25 kg, 125 mg daily; 25 to 35 kg, 187.5 mg daily; over 35 kg, 250 mg daily (which is the adult dose), which is given for 6 weeks.[6] Although the label-indicated duration of therapy is 6 weeks, pulsed dosing regimens and shorter durations have also been reported to be effective.[65,69,70] A dose-response study using terbinafine found that doses >4.5 mg/kg daily were more likely than lower doses to produce a cure in both *Trichophyton* and *Microsporum* infections, with duration of therapy being less important.[71,72] A recent meta-analysis has demonstrated that terbinafine has superior efficacy to griseofulvin for treating tinea capitis because of *Trichophyton* spp.[66]

Itraconazole. Itraconazole continuous and pulse therapies have been used to treat tinea capitis effectively. The dosing regimens used are 5 mg/kg daily for 4 to 8 weeks, or, in the case of pulse therapy, 5 mg/kg daily for 1 week a month, given for 2 to 4 months (see Table 10.5).[65,68] Where the oral solution is used, dosage is reduced to 3 mg/kg daily, whether used continuously or as pulse therapy.[65,73]

Fluconazole. A limited number of studies have shown that continuous fluconazole 6 mg/kg daily lasting 3 weeks can effectively treat tinea capitis (see Table 10.5).[64,65] A comparative study of 5 mg/kg daily for 4 weeks showed similar efficacy to griseofulvin 15 mg/kg daily for 6 weeks.[74] Once weekly therapy with fluconazole for tinea capitis has also been found to be effective.[75,76]

Tinea Corporis, Tinea Cruris, and Tinea Pedis. **Q10.10** Typically, successful treatment is provided by topical antifungal medications; oral antifungals are not generally approved for these indications. Off-label use of oral antifungals may be practical where the tinea involvement is extensive and application of a topical either is not feasible or successful.[42]

Griseofulvin. Griseofulvin is indicated in the United States for the treatment of tinea infections that would not be expected to respond satisfactorily to topical antifungals. Suggested dosage for tinea corporis/cruris is 250 mg twice daily, until a cure is reached.[77] For tinea pedis, the suggested dosage is 660 or 750 mg daily for 4 to 8 weeks.[78]

Terbinafine. Terbinafine has been used for tinea corporis/cruris at 250 mg daily for 2 to 4 weeks, and for tinea pedis at 250 mg daily for 2 to 6 weeks.[77,78]

Itraconazole. A continuous itraconazole regimen of 200 mg daily for 1 week is recommended for tinea corporis/cruris, although a regimen of 100 mg daily for 2 weeks has also been reported to be effective.[77,79,80] For tinea pedis, itraconazole regimens of 100 mg daily for 30 days or 4 weeks, 400 mg daily for 1 week, and 200 mg daily for 2 to 4 weeks have been reported.[42,78,81]

Fluconazole. The suggested dose of fluconazole for tinea corporis/cruris is 150 to 300 mg once weekly, administered for 2 to 4 weeks.[77] The most frequently reported dosage for tinea pedis is 150 mg once weekly, administered for 2 to 6 weeks.[82–85]

Pityriasis (Tinea) Versicolor and Seborrheic Dermatitis. As with dermatophyte infection, the *Malassezia* yeasts associated with tinea versicolor and seborrheic dermatitis are most often successfully treated with topical therapy. However, oral medications are occasionally used off label, particularly when large areas of the body are affected.[42] It should be noted that even though symptoms of tinea versicolor infection may resolve within 2 weeks of therapy, pigmentation abnormalities may persist for many months before returning to normal.[86] Recurrence of tinea versicolor infection is common, particularly in warmer months of the year.[87] Topical corticosteroids (TCS) are commonly used for seborrheic dermatitis, but alternative therapies are commonly used to avoid adverse events associated with prolonged TCS use.[88]

Q10.7 Griseofulvin and terbinafine have not been effective in tinea versicolor infection.[86,87] Based on a meta-analysis, itraconazole therapy (200 mg daily given for 5–7 days or 100 mg daily for 2 weeks) was effective in pityriasis (tinea) versicolor.[89] Fluconazole doses of 300 mg weekly for 1 to 4 weeks have shown high rates of mycologic cure.[42,89]

For seborrheic dermatitis, dosages of oral antifungals used include itraconazole 200 mg daily for 7 days, and terbinafine 250

TABLE 10.5 Dosing for Tinea Capitis in Children

REGIMEN	TERBINAFINE CONTINUOUS		ITRACONAZOLE CONTINUOUS		ITRACONAZOLE PULSE[a]	FLUCONAZOLE CONTINUOUS		FLUCONAZOLE PULSE	GRISEOFULVIN CONTINUOUS	
Formulation	**Oral Granules**	**Tablets**	**Capsules**	**Oral Solution**	**Capsules**	**Tablets**	**Oral Solution**	**Tablets**	**Tablets**	**Oral Solution**
Dose/kg option			5 mg/kg/day	3 mg/kg/day	5 mg/kg/day	6 mg/kg/day	3–6 mg/kg/day	8 mg/kg once weekly	Microsize: 20–25 mg/kg/day	Microsize (125 mg/5 mL): 20–25 mg/kg/day
									Ultramicro-size: 10–15 mg/kg/day	
Duration	6 weeks	2–6 weeks (*T. tonsurans)* 8–12 weeks (*M. canis)*	2–4 weeks (*Trichophyton spp.*) 4–6 weeks (*Microsporum spp.*)	6 weeks	2–4 pulses[a]	3 and 6 weeks	3 weeks	4–8 weeks	≥6 weeks	≥6 weeks
<25 kg	125 mg/day									
25–35 kg	187.5 mg/day									
>35 kg	250 mg/day									
10–20 kg		62.5 mg/day								
21–40 kg		125 mg/day								
>40 kg		250 mg/day								

[a]Itraconazole pulses are 1 week of active treatment followed by 3 weeks off drug.

mg daily for 4 weeks.[42,88,90,91] Patients with facial lesions did not benefit from terbinafine treatment.[91]

Prophylaxis of Tinea Versicolor. A single 400 mg dose of itraconazole once monthly for 6 months may be useful as prophylaxis for tinea versicolor patients who suffer recurrent outbreaks.[86]

Oral Antifungal Use in Children With Superficial Fungal Infection

With the exception of tinea capitis, most superficial dermatophyte infections are infrequently diagnosed in children, and few published data exist on pediatric use of oral antifungals in these other conditions. As a result, oral antifungals are typically not formally approved for use in children; however, pediatric use has been documented widely in the medical literature and has typically demonstrated a safety profile similar to use in adults.[92–94] When using oral antifungals in children, dosing regimens are typically adjusted by weight, as in tinea capitis dosing. Where swallowing the tablets or capsules may be an issue, terbinafine tablets can be crushed or cut up and terbinafine 'granules' can be sprinkled on food; itraconazole capsules can be opened and mixed with fatty foods, such as peanut butter.[95]

Oral suspensions have been developed for many of the oral antifungal medications, which can provide easier dosing for children. These suspensions may have improved pharmacologic qualities and different safety profiles compared with capsule or tablet formulations; therefore, suggested dosing regimens may differ and some formulations may not be used interchangeably.

Children With Onychomycosis. As few children present with distal lateral subungual or proximal subungual onychomycosis (DLSO and PSO, respectively), no oral antifungal has been approved for use in children with dermatophyte onychomycosis. For superficial white onychomycosis or milder cases of DLSO, topical therapy may be preferred over the use of oral antifungals. The choice of therapy should take into account the causative organism, concomitant drug therapies, cost-effectiveness, patient preference, and physician familiarity with the antifungal agents.[95] Mycologic confirmation of infection is suggested before initiation of any oral agent.[95,96]

Terbinafine. Both terbinafine and itraconazole have been reported as effective and safe when used off label for onychomycosis in children.[41,97–101] With terbinafine, the duration of therapy is similar to that in adults: fingernail and toenail onychomycosis for 6 and 12 weeks, respectively.[95] The suggested terbinafine dosage schedule is for patients over 40 kg, 250 mg (= adult dose) daily; 20 to 40 kg, 125 mg daily; and less than 20 kg, 62.5 mg daily.[95]

Itraconazole. For itraconazole capsules, the suggested dosage is 2 and 3 pulses for fingernails and toenails, respectively.[95] Each pulse is 5 mg/kg daily lasting for 1 week, or for those over 50 kg, 200 mg twice daily (= adult dose); 40 to 50 kg, 200 mg daily; 30 to 40 kg, 100 mg daily alternating with 200 mg daily; 20 to 30 kg, 100 mg daily; and 10 to 20 kg, 50 mg on alternating days or 3 times a week. Itraconazole oral solution given as pulse therapy may be another option, at dosages of 3 to 5 mg/kg daily.[95,102]

Fluconazole. Reported use of fluconazole for children with onychomycosis is limited. Use of fluconazole for dermatophyte onychomycosis in general requires relatively long-term therapy. Suggested dosage is an intermittent regimen of 3 to 6 mg/kg once weekly for 18 to 26 and 12 weeks, respectively, for toenail and fingernail onychomycosis.[95,96]

Deep Fungal Infections

Terbinafine. Successful use of terbinafine has been reported in subcutaneous and systemic mycoses, such as chromoblastomycosis, sporotrichosis, fungal mycetoma, aspergillosis, and histoplasmosis.[103] Terbinafine regimens are not clearly established; however, effective dosing for systemic mycoses would appear to require 500 to 1000 mg daily.[103]

Itraconazole. Itraconazole capsules are indicated for the treatment of the following fungal infections: blastomycosis (pulmonary and extrapulmonary), histoplasmosis (including chronic cavitary, pulmonary, and disseminated, nonmeningeal disease), and aspergillosis (pulmonary and extrapulmonary in patients who are intolerant of or refractory to amphotericin B therapy).[1] Life-threatening histoplasmosis and blastomycosis may require intravenous itraconazole.[104]

Fluconazole. Fluconazole is effective for systemic *Candida* infections, including candidemia and disseminated candidiasis, and is available as an intravenous formulation.[4,104] High-dose oral fluconazole (400–600 mg daily) has been recommended for coccidioidal meningitis; however, fluconazole has relatively poor efficacy against endemic mycoses, such as histoplasmosis, blastomycosis, and paracoccidioidomycosis.[104,105] The high oral bioavailability of fluconazole is an asset, but the narrow spectrum of action limits its use as a prophylactic.[104]

Other Off-Label Uses for Oral Antifungals

Terbinafine. There have been isolated reports of terbinafine use in patients with Majocchi's granuloma,[106] tinea imbricata,[107,108] cutaneous sporotrichosis,[109] black piedra,[110] aspergillosis,[111] and chromoblastomycosis.[112]

Itraconazole. Itraconazole may also be effective in the treatment of Majocchi's granuloma,[34] HIV-associated eosinophilic folliculitis,[113] tinea imbricata,[107] vaginal candidiasis, chronic mucocutaneous candidiasis,[62,114] cutaneous sporotrichosis,[109] and other *C. albicans* infections.[61]

Fluconazole. Fluconazole is effective in the treatment of cutaneous candidiasis, using 150 mg once weekly given for 2 to 4 weeks.[82,115]

Griseofulvin. Griseofluvin, 500 mg twice daily for 4 to 6 weeks, has been effective treatment for tinea imbricata, but may require concomitant topical therapy.[108]

Contraindications

All of the oral antifungals are contraindicated for patients who are allergic to the drug or its excipients. See Risks Profiles for terbinafine, itraconazole, and fluconazole for a succinct summary (Boxes 10.1 to 10.3).

Terbinafine. Terbinafine (capsule or granule formulation) is not recommended for patients with chronic or active liver disease.[5,6] Terbinafine clearance is reduced by approximately 50% in patients with renal impairment (creatinine clearance <50 mL/min) compared with normal volunteers, and terbinafine may not be a suitable choice for patients with renal impairment.[5] Terbinafine, as a CYP2D6 inhibitor, also shows potential drug interactions with a number of other drug classes sharing the CYP2D6 pathway (Table 10.6).[5,6]

Itraconazole, fluconazole and the newer triazoles. Q10.11 Itraconazole should not be administered for onychomycosis in patients with evidence of ventricular dysfunction, such as congestive heart failure or history of congestive heart failure, as temporarily decreased cardiac contractility has been noted in a healthy volunteer study.[1–3] Similarly, albeit not always strictly contraindicated, caution is advised when using any of the azoles in patients with potentially proarrhythmic conditions.[4,10–12]

• BOX 10.1 Drug Risks Profile—Terbinafine

Contraindications
Hypersensitivity to terbinafine or components of formulation

Chronic or active hepatic disease

Boxed Warnings
None listed

Warnings & Precautions[a]
Hepatic
[a]Cases of hepatic failure, leading to transplant, death reported

Hematologic
[a]Lymphocytopenia, severe neutropenia—reversible
Thrombotic microangiopathy, hemolytic uremic syndrome rarely

Appropriate Use
[a]Because of potential toxicity, dx confirmation skin nails required

Nasal & Oral
[a]Smell and/or taste disturbance

Hypersensitivity Reactions
[a]Cutaneous (esp. SCLE) and systemic lupus erythematosus
Allylamine sensitivity, including cross-reactions

Ocular
Rare changes in lens, retina reported

Psychiatric
Rare cases of depression reported

Pregnancy Prescribing Status
Traditional US Food and Drug Administration rating—category B

Newer rating[b]—low risk

[a]Under "Warnings & Precautions" these adverse effects can be considered relatively high risk or important clinical scenarios to avoid.
[b]See Chapter 65, Dermatologic Drugs During Pregnancy and Lactation, for detailed explanations of terms for "Newer rating" based on 2015 US Food and Drug Administration rulings.
SCLE, subacute cutaneous lupus erythematosus.
Data from Facts & Comparisons eAnswers [online database]. St. Louis: Wolters Kluwer. [https://www.wolterskluwercdi.com/facts-comparisons-online/].

• BOX 10.2 Drug Risks Profile—Itraconazole

Contraindications
Hypersensitivity to itraconazole or components formulation

Numerous drug interactions (see Table 10.7)

Boxed Warnings
Congestive heart failure

Numerous drug interactions (see Table 10.7)

Warnings & Precautions[a]
Hepatic
[a]Rare cases serious hepatotoxicity, including liver failure, death

Neurologic
Transient or permanent hearing loss, esp. when coadminister quinidine
Neuropathy (no additional details given)

Hypersensitivity Reactions
[a]No information on cross-reaction with other azoles

Cardiovascular
[a]May induce CHF, esp. doses at least 400 mg daily
[a]Caution with risk factors CHF, including ischemic/valvular heart disease, significant pulmonary disease, renal failure

Hematologic
[a]Caution severely neutropenic patients

Pregnancy Prescribing Status
Traditional US Food and Drug Administration rating—category C

Newer rating[b]—low risk

[a]Under "Warnings & Precautions" these adverse effects can be considered relatively high risk or important clinical scenarios to avoid.
[b]See Chapter 65, Dermatologic Drugs During Pregnancy and Lactation, for detailed explanations of terms for "Newer rating" based on 2015 US Food and Drug Administration rulings.
CHF, Congestive heart failure.
Data from Facts & Comparisons eAnswers [online database]. St. Louis: Wolters Kluwer. [https://www.wolterskluwercdi.com/facts-comparisons-online/].

• BOX 10.3 Drug Risks Profile—Fluconazole

Contraindcations
Hypersensitivity to fluconazole or components of formulation

Drug interactions involving QT prolongation

Boxed Warnings
None listed

Warnings & Precautions[a]
Hepatic
[a]Serious hepatic toxicity, including liver failure, death

Hypersensitivity Reactions
Rare exfoliative skin reactions, some fatal with serious comorbidities

Cardiovascular
[a]QT prolongation with ventricular dysrhythmias, usually in combination drugs known to prolong QT interval

Pregnancy Prescribing Status
Traditional US Food and Drug Administration rating—unrated

Newer rating[b]—**Moderate-high risk** (single dose—low risk)

[a]Under "Warnings & Precautions" these adverse effects can be considered relatively high risk or important clinical scenarios to avoid.
[b]See Chapter 65, Dermatologic Drugs During Pregnancy and Lactation, for detailed explanations of terms for "Newer rating" based on 2015 US Food and Drug Administration rulings.
Data from Facts & Comparisons eAnswers [online database]. St. Louis: Wolters Kluwer. [https://www.wolterskluwercdi.com/facts-comparisons-online/].

For both itraconazole and fluconazole, serious cardiovascular events, such as prolongation of the QT interval, torsades de pointes, ventricular tachycardia, cardiac arrest, and sudden death have been noted. Coadministration of cisapride, pimozide, and quinidine, which are known to prolong the QT interval, is contraindicated with all the azoles.[1–4,10–12] Itraconazole is also contraindicated with dofetilide, levacetylmethadol (levomethadyl), the HMG CoA-reductase inhibitors, such as lovastatin and simvastatin, the benzodiazepines midazolam and triazolam, the calcium-channel blocker, nisoldipine, and CYP3A4 ergot alkaloids, such as dihydroergotamine, ergometrine (ergonovine), ergotamine and methylergometrine (methylergonovine).[1–3] Fluconazole is contraindicated with erythromycin, cisapride, astemizole, pimozide, and quinidine, as well as terfenadine, when dosing of fluconazole is 400 mg or greater.[4] Itraconazole and posaconazole inhibit the CYP3A4 enzyme; fluconazole and voriconazole inhibit CYP3A4 and CYP2C9, with voriconazole also involved with CYP2C19, giving these two drugs somewhat expanded interactions compared with itraconazole.[1–4,10–12] Fluconazole and the newer triazoles have not been studied in all medications contraindicated with itraconazole, but interaction may occur between these azoles and the itraconazole-contraindicated medications.[4,10–12]

Itraconazole use is strongly discouraged in patients with active liver disease, elevated or abnormal liver enzymes, or previous experience of liver toxicity with other drugs.[1–3] Caution is advised when any of the azoles are used in these patient populations, and monitoring of liver functions is suggested for all patients both before and during therapy.[4,10–12]

Caution should also be used when administering an azole agent to patients who have exhibited past sensitivity to another azole agent, as there may be some potential for cross-reaction between azole drugs.[1–4,10–12]

TABLE 10.6 Drug Interactions—Terbinafine

Drug Category	Drug Examples	Comments
Relatively High-Risk Drug Interactions[a]		
Narcotics	All members	CYP2D6 substrates → ↑ drug levels and potential toxicity in conjunction with terbinafine which is strong CYP2D6 inhibitor (codeine → morphine conversion depends activity of CYP2D6)
β-blockers	***Propranolol, metoprolol, timolol***	*same*
Tricyclic antidepressants	***Doxepin***, ***amitriptyline***	*same* (also clomipramine, desipramine)
Antipsychotics	***Pimozide***	*same* (also risperidone, haloperidol, clozapine; pimozide significant risk QT prolongation)
Antidysrhythmics	***Encainide***, ***flecainide***, ***mexiletine***	same
SSRI antidepressants	Fluoxetine, paroxetine	*same*
NSRI antidepressants	Trazadone, venlafaxine	*same*
Antidysrhythmics	***Amiodarone***	Stronger CYP2D6 inhibitor → ↑ terbinafine (a CYP2D6 substrate) drug level and potential toxicity
Tricyclic antidepressants	Clomipramine, desipramine	*same*
SSRI antidepressants	Fluoxetine, paroxetine (>> sertraline)	same (common combinations with doxepin, amitriptyline; potentially with terbinafine as well)
NSRI antidepressants	Bupropion	*same*
HIV-1 protease inhibitor	Ritonavir	*same*
Antipsychotics	Haloperidol, thioridazine	*same*
Lower-Risk Drug Interactions		
Anticonvulsants	*none*	No known CYP2D6 inducers (members of these 3 groups inducers of other major CYP isoforms)
Rifamycins	*none*	*same*
Antifungals	*none*	*same*
NSAID	Celecoxib	Weaker CYP2D6 inhibitors
Antimalarials (4-aminoquinoline structure)	Hydroxychloroquine	*same*
H_2 antihistamines	Cimetidine	*same*

[a]Overall highest-risk drug interactions indicated in ***bold italics***.
CYP, Cytochrome P-450; *HIV*, human immunodeficiency virus; *NSAID*, nonsteroidal anti-inflammatory drugs; *NSRI*, nonselective reuptake inhibitor; *SSRI*, selective serotonin reuptake inhibitor.
The dramatic increase in number of drug interactions in medicine requires some degree of selectivity in these tables (common usage, relative risk, focus on outpatient rx)

Data from Facts & Comparisons eAnswers (online database). St. Louis: Wolters Kluwer. (https://www.wolterskluwercdi.com/facts-comparisons-online/); Hansten PD, Horn JR. The Top 100 Drug Interactions: A Guide to Patient Management, 2019 Edition. Freeland, WA: H&H Publications, 2019. (http://www.hanstenandhorn.com/).

Adverse Effects

Terbinafine. Terbinafine is considered a safe medication for the general population, as well as children, the elderly, transplant patients, diabetics, and HIV patients.[116] The safety profile of the oral granule formulation is comparable to that of the tablet formulation (see Table 10.6).[5,6] The more common adverse effects (AE) with terbinafine use are headache, gastrointestinal symptoms (diarrhea, dyspepsia, abdominal pain, nausea, and flatulence), dermatologic manifestations (rash, pruritus, urticaria), liver enzyme abnormalities greater than or equal to twice the upper limit of normal, taste disturbance, and visual disturbance.[5] Events were noted in small proportions of subjects and were generally mild, and transient.

Rare severe reactions have been noted with terbinafine. **Q10.12** Severe skin reactions, such as erythema multiforme, toxic epidermal necrolysis, and Stevens–Johnson syndrome, were reported with oral terbinafine use in 38, four, and nine cases, respectively, out of 2313 adverse reactions reported in the World Health Organization worldwide database (data reported December 1996).[117] Severe skin reactions may present initially as a serum sickness-like reaction.[117,118]

Rare development of idiosyncratic hepatobiliary dysfunction has been reported (1:45,000–1:120,000).[119,120] Hepatitis produced by terbinafine typically develops within 4 to 6 weeks of treatment initiation, and has the features of both hepatocellular necrosis and cholestatic injury.[120] In most cases, liver functions return to normal several months after stopping the medication.[120]

Terbinafine has been reported infrequently to precipitate or exacerbate cutaneous and subacute cutaneous lupus erythematosus, and should be discontinued in patients showing signs of lupus erythematosus.[5] **Q10.11** Postmarketing data also revealed several cases of subjects experiencing depressive symptoms during use of terbinafine.[5] Terbinafine is rated as pregnancy category B and is not recommended for routine use in pregnant or nursing women. No effects on testosterone levels were detected with terbinafine use in a healthy male population.[121]

Itraconazole. With itraconazole, the common AE are headache, gastrointestinal disorders, and cutaneous disorders (Table 10.7).[1–3] When itraconazole is given as a pulse regimen for the treatment of onychomycosis, it may be associated with an improved AE profile compared with the continuous regimen using this triazole.[122]

Q10.12 In the General Practice Research Database in the United Kingdom sampling nearly 27,000 patients, two cases of serious skin disorders were found with itraconazole use: angioedema (one case) and erythema multiforme (one case).[123] Stevens–Johnson syndrome has been rarely reported with itraconazole.[1]

Abnormal liver function tests were found in 3% of 1845 patients treated with continuous itraconazole compared with 1.9% of 2867 patients treated with pulse itraconazole (200 mg twice daily, for 1 week out of each month).[122] Serious adverse liver events were recorded in 3.2 per 100,000 prescriptions.[120] The estimated incidence of itraconazole inducing clinically significant symptoms and signs of hepatobiliary dysfunction, for which no other cause was apparent, is 1:500,000.[119]

Itraconazole is rated as pregnancy category C and is not to be used in women who are pregnant, planning a pregnancy, or nursing. In contrast to ketoconazole, use of itraconazole showed no effect on androgen levels and that alteration of male reproduction is unlikely.[124,125]

Safety profiles are similar for capsules as the oral suspension.[1,2] Preclinical testing of the hydroxy-propyl-β-cyclodextrin vehicle used in the itraconazole oral suspension found some potential for pancreatic adenocarcinomas in rats, but no other tested animal species.[2] The potential for carcinoma development in humans has not been determined.

Fluconazole. Both at the doses used in dermatology and at higher doses, fluconazole has shown a favorable AE profile.[4,126–128] AE profiles seen in pediatric fluconazole use are similar to those in adults.[4] The most frequently reported AE in clinical trial patients receiving these azoles are headache, nausea, vomiting, abdominal pain, and diarrhea (see Table 10.7).[4,10–12,120,123,126–128]

Q10.12 Cases of toxic epidermal necrolysis, Stevens–Johnson syndrome, angioedema, and erythema multiforme have been reported with fluconazole.[4,10,120,123,128] Rare cases of serious hepatic toxicity have been reported, with no obvious relationship to daily dose, therapy duration or other factors.[4] Self-limiting hepatic and biliary abnormalities were noted in 0.5% of patients using fluconazole, whereas elevated enzyme levels, particularly aspartate aminotransferase (AST), occurred in 10% of subjects using chronic fluconazole therapy.[120]

Fluconazole is rated as pregnancy category D, with birth defects being noted by a few case reports in subjects using a high doses (400–800 mg/day). Fluconazole, should not be used in women who are pregnant, planning a pregnancy, or nursing. In contrast to ketoconazole, fluconazole at 25 to 50 mg/day showed no significant effect on testosterone levels in healthy male volunteers.[129]

Voriconazole. It is of historical interest that voriconazole-induced photosensitivity reports have followed by squamous cell carcinoma and melanoma.[10] Furthermore, a variety of visual disturbance have been reported in clinical trials and postmarketing surveillance.[10]

Drug Interactions

In general, before prescribing a new drug, it is important to obtain a complete history of all drugs the patient is currently taking, both prescription and nonprescription. The inquiry should extend to herbal and recreational agents.

Q10.13 The potential drug interactions associated with terbinafine are listed in Table 10.6.[5,6] Terbinafine has relatively few drug interactions compared with the azoles. Terbinafine has been reported to inhibit CYP2D6.[5,6] Clinicians should be cautious regarding concomitant administration of terbinafine with CYP2D6 substrates, such as the tricyclic antidepressants doxepin and amitriptyline.

The azole drugs share many important drug interactions (see Table 10.7). Coadministration of itraconazole (capsules, injection, or oral solution) with cisapride, pimozide, quinidine, dofetilide, or levacetylmethadol (levomethadyl) is contraindicated.[1] **Q10.14** Similarly, concomitant use of fluconazole with cisapride, or the antihistamines terfenadine (fluconazole doses ≥400 mg) and astemizole is contraindicated.[4] Levels of these drugs may become elevated and cause serious cardiovascular events, such as QT prolongation, torsades de pointes, ventricular tachycardia, cardiac arrest, and/or sudden death.[1] Although terfenadine, astemizole, and cisapride are no longer available in many countries, these contraindications may still be relevant in others.

Q10.13 The azole drugs interfere to varying degrees with CYP3A4, with fluconazole also inhibiting CYP2C9. Drugs using these metabolic pathways may have drug concentrations altered when given concomitantly with an azole antifungal agent. Where significant interaction has been reported, concomitant drug levels or activities may need to be monitored and/or dose reduced, to minimize the interaction risk (see Table 10.7). **Q10.14** The most notable CYP2C9 interactions with the azoles are through coadministration of fluconazole with warfarin, which can lead to significant increases in international normalized ratio (INR) values and excessive anticoagulation. When itraconazole (and to a degree fluconazole of at least 300 mg daily) and cyclosporine are given concomitantly, careful monitoring of cyclosporine concentration and serum creatinine concentration is recommended.[4] Blood glucose levels may require careful monitoring when oral hypoglycemic agents are used concomitantly with azoles. Similarly, agents with narrow therapeutic windows, such as phenytoin and theophylline, may need to have careful monitoring of drug levels, as interactions secondary to azole use may more easily predispose the patient to significant AE.

Monitoring Guidelines

Liver function tests should be performed in all patients before prescribing terbinafine and azoles for courses of at least 3 to 4 weeks continuously (serum transaminase tests: alanine aminotransferase [ALT], AST).[5] Patients should be instructed to report any symptoms of liver dysfunction, such as persistent nausea, anorexia, fatigue, vomiting, right upper abdominal pain or jaundice, dark urine, or pale stools. Patients reporting such symptoms, or otherwise suspected of having hepatic dysfunction, should discontinue terbinafine and have a complete liver profile performed.[5] The US package insert indicates that physicians should consider

TABLE 10.7 Drug Interactions—Itraconazole and Fluconazole

Drug Category	Drug Examples	Comments
Relatively High-Risk Drug Interactions[a] CYP3A4 Inhibition (itraconazole >> fluconazole *unless dose over 200 mg daily*)		
Calcineurin inhibitors	***Cyclosporine*** (and oral ***tacrolimus***)	↑ drug levels and risk of severe toxicity (HBP, renal failure, hyperlipidemia and others)
Statins	***Simvastatin, atorvastatin, lovastatin***	*same* (risk myopathy, rhabdomyolysis)
Benzodiazepines	***Alprazolam, triazolam, midazolam***	*same* (risk excessive sedation)
Antipsychotics	***Pimozide***	*same* (risk QT prolongation, torsades de pointes)
Myeloperoxidase inhibitors	***Dapsone***	*same* (risk hemolysis, agranulocytosis)
Anticoagulants	***Warfarin***	R-enantiomer of warfarin (2nd most important substrate pathway after CYP1A2) risk bleeding
Antigout meds	***Colchicine***	Possible ↑ myelotoxicity
Relatively High-Risk Drug Interactions[a] CYP2C9 Inhibition (primarily fluconazole)		
Anticoagulants	***Warfarin***	S-enantiomer of warfarin major substrate pathway (most important) severe risk bleeding
Statins	***Fluvastatin, rosuvastatin***	Statins metabolized by CYP2C9 pathway; risk myopathy, rhabdomyolysis
Antidepressants	***Doxepin***	Main antidepressant metabolized by CYP2C9; risk excessive sedation, QT prolongation → torsades
Anticonvulsants	***Phenytoin, valproic acid***	↑ drug levels and risk of severe toxicity
Lower-Risk Drug Interactions—CYP3A4 Induction (only itraconazole CYP3A4 substrate)		
Rifamycin antibacterials	Rifampin, rifabutin, rifapentine	CYP3A4 ***induction*** with resultant ↓ itraconazole drug levels, loss efficacy; this effect begins 1–2 wk
Aromatic anticonvulsants	Phenytoin, carbamazepine, phenobarbital	*same* (also nonaromatics, such as primidone, ethosuximide)
Antifungals	Griseofulvin	Weak CYP3A4 inducer
Lower-Risk Drug Interactions—CYP3A4 Inhibition (itraconazole >> fluconazole *unless dose over 200 mg daily*)		
H_1 antihistamines	Fexofenadine, loratadine	Primary antihistamines (1st or 2nd generation), which are CYP3A4 substrates; historically, terfenadine, astemizole (both off the market) induced torsades de pointes in combination CYP3A4 inhibitors)
Hormonal	Estrogens, combined oral contraceptives	In general, modest risk (conceptually venous thromboembolism risk; likely a stronger genetic risk)
Retinoids	Isotretinoin (possible acitretin)	In general, modest risk (most of very few interactions other mechanisms)
Lower-Risk Drug Interactions—CYP2C9 Inhibition (primarily fluconazole)		
Antibacterials	Sulfonamides	In general, modest risk (likely genetic factors > dose with SJS/TEN risk)
Statins	Pitavastatin	Statin metabolized by CYP2C9 pathway to a smaller extent; risk myopathy, rhabdomyolysis
NSAID	Diclofenac, ibuprofen, piroxicam	In general, modest risk (conceptually ↑ risk GI and renal AE/SAE)

[a]Overall highest-risk drug interactions indicated in ***bold italics***.

AE/SAE; Adverse effects/serious adverse effects; *CYP,* cytochrome P-450; *GI*, gastrointestinal; *HBP*, high blood pressure; *SJS/TEN*, Stevens–Johnson syndrome/toxic epidermal necrolysis.

The dramatic increase in number of drug interactions in medicine requires some degree of selectivity in these tables (common usage, relative risk, focus on outpatient rx)

Data from Facts & Comparisons eAnswers (online database). St. Louis: Wolters Kluwer. (https://www.wolterskluwercdi.com/facts-comparisons-online/); Hansten PD, Horn JR. The Top 100 Drug Interactions: A Guide to Patient Management, 2019 Edition. Freeland, WA: H&H Publications, 2019. (http://www.hanstenandhorn.com/).

monitoring complete blood counts in patients with known or suspected immunodeficiency, who are administered oral terbinafine for longer than 6 weeks.[5] **Q10.11** Patients should also be instructed to report taste disturbance, depression symptoms, and progressive rash.[5]

When giving azoles, LFT monitoring should be considered for any subject.[1] LFT should be done for any subject with preexisting hepatic function abnormalities, or any subject with previous experience of liver toxicity with other medications.[1]

During prolonged griseofulvin therapy, periodic assessment of renal, hepatic, and hematopoietic functions should be performed.[9]

Conclusion

Systemic oral antifungal agents have widespread use in dermatology. Dosing and efficacy vary with specific indications. Clinical use has demonstrated the general safety of oral antifungal use. However, possible risks exist for each agent, and careful monitoring of the patient should be undertaken regardless of agent used.